

Arab Academy for Science, Technology & Maritime Transport

College of engineering

Mechatronics engineering department

semester: 9

Graduation Project Proposal

Project title: Design & implementation of active ankle prosthesis

Professors Names: Dr. Ahmed El Sawaf; Dr. Mostafa Fouz

Number of students : four

Students Qualifications

- Experience In electronics,electrical wiring and using microcontrollers.
- Programming skills In C, python and arduino IDE (C,C++).
- Familiarity with Matlab & Simulink .
- Experience in computer aided design using Autocad, Inventor and solidworks.
- Knowledge in embedded systems.

Statement of problem

- The problem statement for our graduation project is that there are many people who lost their legs or parts of their legs in accidents or during armed disputes or other medical conditions like diabetic patients or chronic limb ischemia in atherosclerosis or infectious diseases. Lower limb amputations constitute a severe health problem, causing high rates of morbidity, mortality and considerable social impact. Amputees in the Middle East struggle to afford prosthetic limbs in general since most manufacturers are outside of the region the prices of prosthetic limbs cost more because of customs and other taxes. Therefore amputees in the middle east struggle to lead a normal life and do regular activities like walking running ,etc.. Which could cause them even more mental pain.
- Based on the national limb resource center and world health organization, 2.1 million people are living with limb loss in the USA, and that number is expected to double by 2050, around the world there are nearly 185,000 people have amputation every year which means that are from 300 to 500 amputation

surgeries each day, in Afghanistan and Iraq there are nearly 1,558 military people lost their limbs, and the number of amputations caused by diabetes increased by 24% from 1988 to 2009.
Objectives Our objective to decrease the disability of amputees, to help them lead a normal and a healthy life and be productive in the community. So we decided to make the active ankle prosthesis instead of using a passive ankle prosthesis, where in the passive there is very little assistance for the amputee in doing regular tasks like taking the stairs, walking and even running causing muscle exhaustion and make regular tasks very hard. The target of this project is to make active ankle prosthesis more affordable and available for amputees to buy and lead a normal life.
Project Idea We are going to design an active ankle prosthesis, it will be light in weight, high autonomy, energy efficient system, so the batteries last longer, support various age groups and various patient weight classes.
Technical approach • Feasibility study and market research targeting the needs of current amputee's In the MENA region to determine the requirements of the ankle. • Comprehensive literature review to learn about previous research in active ankle prosthesis. • Design of the mechanical system in parallel with the electrical system including the control of the active ankle. • Simulation of the active ankle using MATLAB and Simulink. • Manufacturing the ankle based on the optimized design acquired by simulation. • Integrating the mechanical and electrical system. • Verification of the objectives required and objectives delivered of the project, then testing. • Comparing simulation and experimental results
Deliverables

- Simulation and modeling results.
- A cost efficient working prototype.
- Passing the standardized tests of ankle prosthesis in the MENA region to be able to test the prototype on patients.
- A working prototype of an active ankle which can be available in the market in the MENA region.